Question 1

In a certain tumor genome-wide study, the researchers divided the genome into many windows of equal length and counted the number of somatic SNVs (single nucleotide variations) in each window. It is known that in a certain type of sample, mutations occur approximately independently and randomly on the genome, and the number of mutations in each 20 kb window follows a Poisson distribution.
It is known that the average mutation rate of this tumor sample is 120 mutations per Mb.
Let the random variable X represent the number of mutations in a randomly selected 20 kb window.
(1)Calculate the probability of exactly four mutations occurring within this 20 kb window.
(2)Calculate the probability that there are fewer than 3 mutations in this 20 kb window.
(3)The researchers defined the "mutation enrichment window" as: a 20 kb window with more than 5 mutations. Calculate the probability that any randomly selected 20 kb window becomes a "mutation enrichment window".

Reference solution
Step 1: Determine the parameters of the Poisson distribution
The average mutation rate is 120 times per Mb, and
20 kb = 0.02 Mb
So the expected number of mutations in this window is
$\lambda = 120 \times 0.02 = 2.4$
Therefore

$X \sim \text{Poisson}(2.4)$
The probability formula of the Poisson distribution is

$$P(X = k) = \frac{e^{-\lambda}\lambda^k}{k!}$$

here
$\lambda = 2.4$.
The probability of exactly 4 mutations occurring
$P(X=4) \approx 0.1254$

(2) Probability of less than 3 mutations
"Less than 3" means $P(X<3)=P(X=0)+P(X=1)+P(X=2)$

Calculate separately:
$P(X=0) = 0.09072$
$P(X=1) \approx 0.09072 \times 2.4 = 0.2177$
$P(X=2) \approx 0.2613$

Therefore

P(X<3) = 0.09072 + 0.2177 + 0.2613 = 0.5697
(3) Probability of more than 5 mutations
"More than 5" means P(X>5)=1−P(X≤5)

P(X=0) ≈ 0.09072
P(X=1) ≈ 0.2177
P(X=2) ≈ 0.2613
Continue to calculate:
P(X=3) ≈ 0.2090
P(X=4) ≈ 0.1254
P(X=5) ≈ 0.0602
So
P(X≤5) ≈ 0.09072 + 0.2177 + 0.2613 + 0.2090 + 0.1254 + 0.0602 ≈ 0.9643
Therefore
P(X > 5) = 1 - 0.9643 = 0.0357

Question 2

2. In a biological tracer experiment, a radioactive isotope is used to label molecules inside cells. Let X denote the lifetime of a radioactive atom before it decays. Suppose X follows an exponential distribution with rate parameter $\lambda = 0.2\ \mathrm{min}^{-1}$.
Answer the following questions:
1）What is the half-life $\mathrm{t}_{1/2}$, defined as the time by which 50% of the radioactive atoms have decayed?
2）What is the probability that a randomly chosen radioactive atom remains undecayed for more than 10 minutes？
3）Suppose an atom has already remained undecayed for 5 minutes. For such an atom, is the probability that it will remain undecayed for more than 10 additional minutes from that moment larger than, smaller than, or unchanged compared with the probability at time 0 ?

Reference solution

1）The half-life satisfies $\mathrm{P}\left(X \le t_{1/2}\right) = 0.5$, so $1 - \mathrm{e}^{-\lambda \mathrm{t}_{1/2}} = 0.5$

$e^{-0.2\, t_{1/2}} = 0.5$, *hence* $t_{1/2} = \frac{1n2}{0.2} \approx 3.47$ *min*

The half-life is about 3.47 minutes.
2) For an exponential distribution,$\mathrm{P}(\mathrm{X} > \mathrm{t}) = \mathrm{e}^{-\lambda \mathrm{t}}$

so $\mathrm{P}(\mathrm{X} > 10) = \mathrm{e}^{-0.2\times 10} = \mathrm{e}^{-2} \approx 0.1353$

The probability is about 0.1353, or 13.53%.

3) The exponential distribution is memoryless, so surviving the first 5 minutes does not change the distribution of the remaining lifetime.

$\mathrm{P}(\mathrm{X} > 15 \mid \mathrm{X} > 5) = \mathrm{P}(\mathrm{X} > 10) = \mathrm{e}^{-2} \approx 0.1353$

The probability is unchanged.

Question 3

Question3 is about normal distribution.

（1） What is normal distribution？What is its probability density function？

（2） For a random variable following a normal distribution $N(\mu, \sigma^2)$, how does increasing or decreasing μtransform the graph of the distribution? How does increasing or decreasing σaffect the graph?

（3） What is the standard normal distribution？In the standardization transformation, how is the z-score defined?

（4） Suppose the weights of a batch of eggs follow a normal distribution

$$X \sim N(50\ \mathrm{g}, 5^2).$$

What is the z-score of an egg that weighs 45g？

（5） What is the probability that one egg weighs less than 42.3g？

（6） Suppose I randomly select one egg from the eggs weighing at least 50g. What is the probability that the selected egg weighs at least 55g？

Reference solution

1）A normal distribution is a continuous probability distribution that is symmetric about its mean. It is fully determined by two parameters: the mean $\mu$and the variance σ^2.

Its probability density function is

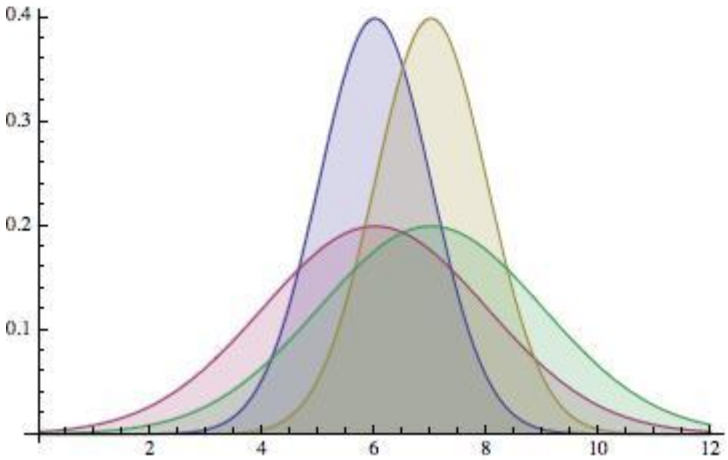
$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(0-\frac{(x-\mu)^2}{2\sigma^2}\right), -\infty < x < \infty, \sigma > 0$$

2）Increasing μshifts the graph to the right.

Decreasing μshifts the graph to the left.

Increasing σmakes the graph wider and flatter.

Decreasing σmakes the graph narrower and taller.



3）The standard normal distribution is a normal distribution with mean $0$and variance 1, denoted by

$$Z \sim N(0,1).$$

The z-score in the standardization transformation is defined as

$$z = \frac{x-\mu}{\sigma}.$$

This transformation converts a normal random variable $X \sim N(\mu, \sigma^2)$into a standard normal random variable $Z \sim N(0,1)$.

4）For $x = 45\ \mu = 50\ \sigma = 5$

$$z = \frac{45-50}{5} = \frac{-5}{5} = -1$$

So, the z-score of the egg is -1.

5）

$$P(X < 42.3) = P\left(Z < \frac{42.3-50}{5}\right) = P(Z < -1.54) \approx 0.0618$$

So, the probability is approximately 0.0618, or 6.18%.

6）Since

$$P(X \geq 55) = P\left(Z \geq \frac{55-50}{5}\right) = P(Z \geq 1) = 0.1587$$

and

$$P(X \geq 50) = 0.5,$$

we get

$$P(X \geq 55 \mid X \geq 50) = \frac{0.1587}{0.5} = 0.3174.$$

So, the probability is approximately 0.3174, or 31.74%.

Question 4:

In single-cell RNA sequencing (scRNA-seq) experiments, researchers often face a trade-off between sample size (number of cells sequenced) and experimental cost. Sample size directly affects the accuracy and stability of gene expression estimates. A lab is studying the expression levels of a marker gene in immune cells. Assuming: the gene expression level approximately follows a normal distribution with mean μ = 25 TPM and standard deviation σ = 8 TPM.

Three team members use different sequencing strategies: Mark sequences 6 cells per sample, Marina sequences 300 cells per sample, and Piccolo sequences 2000 cells per sample. Each member repeats their sampling process 500 times (with replacement) to evaluate the stability of their estimates. **Using R**, simulate the **sampling** processes for the three members from the normal population. Plot the distribution of their sample means (histogram and fit line), calculate the standard deviation of the **sample means** obtained from the sampling, and compare it with the theoretical value of the standard error.

Hint: Use `rnorm(n, mean, sd)` and `hist()` functions.

Reference solution

```
#parameter
mu <- 25
sigma <- 8
n_rep <- 500
sample_size <- c(6, 300, 2000)

set.seed(123)

# calculate mean and sd of mean distribution
# sample and get mean value: lappy for list
sample_means_list <- lapply(sample_size, function(n) {
  replicate(n_rep, mean(rnorm(n, mean = mu, sd = sigma)))
})
names(sample_means_list) <- paste0("n=", sample_size)

# sapply: list in and the simplest format out(here: vector)
sample_sigma <- sapply(sample_means_list, sd)

##Due to the Central Limit Theorem (CLT)
theoretical_sigma <- sigma / sqrt(sample_size)
# print result
print(cbind(n = sample_size,
         theoretical_sigma = theoretical_sigma,
```

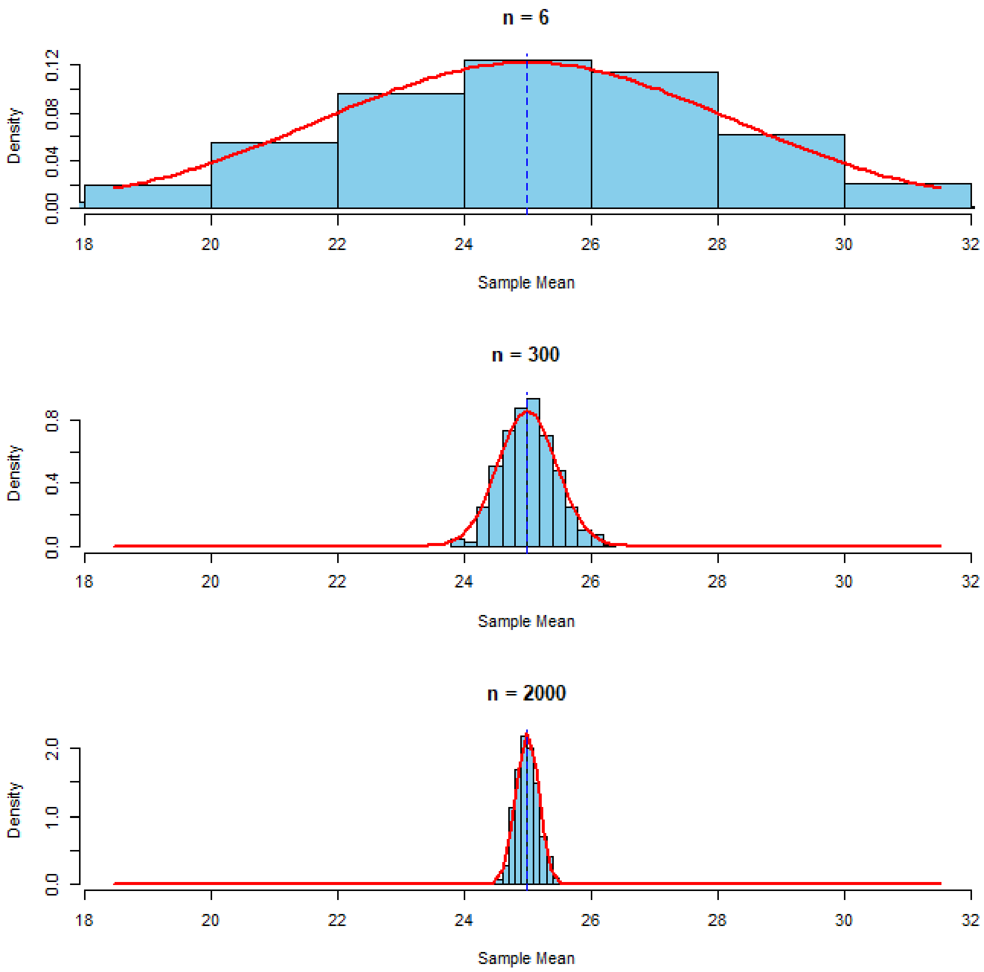
```r
        real_sigma = sample_sigma))

#plot
# xlim(optional) easier to compare
xlim_range <- mu + c(-2, 2) * max(sigma / sqrt(sample_size))

par(mfrow = c(3,1))
for (i in seq_along(sample_size)) {
  hist(sample_means_list[[i]], prob = TRUE,
         main = paste("n =", sample_size[i]),
         xlab = "Sample Mean", col = "skyblue",
         xlim = xlim_range)

  #optionl(plot the real distribution)
  curve(dnorm(x, mu, sigma/sqrt(sample_size[i])),
          add = TRUE, col = "red", lwd = 2)
  abline(v = mu, col = "blue", lty = 2)
}
```

```
           n theoretical_sigma real_sigma
n=6        6         3.2659863  3.0695792
n=300    300         0.4618802  0.4230787
n=2000  2000         0.1788854  0.1753116
```



Question 5

An agricultural research institute is developing a new compound fertilizer aimed at increasing corn yield. Based on extensive long-term studies and field trial data of similar fertilizers, the population standard deviation of the increase in corn yield per acre caused by such fertilizers is known to be $\sigma=18.2$kg/acre.

To evaluate the actual effect of this new fertilizer, researchers applied it to 50 experimental plots. After one growing season, the increase in corn yield for these 50 plots was measured. The sample mean increase in corn yield per mu was found to be $\bar{x}=75.6$kg/acre.

Based on this data, calculate the 95% confidence interval for the population mean increase in corn yield caused by this new fertilizer.

Reference solution

$$\bar{x} = 75.6\ 公斤/亩$$

$$\sigma = 18.2\ 公斤/亩$$

$$n = 50$$

$$\alpha = 1 - 0.95 = 0.05$$

$$置信区间 = \bar{x} \pm Z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}$$

$$Z_{\alpha/2} = \mathrm{Z}_{0.025} \approx 1.96$$

$$Z_{0.025} \cdot \frac{\sigma}{\sqrt{n}} = 1.96 \cdot \frac{18.2}{\sqrt{50}} = 1.96 \cdot \frac{18.2}{7.0711} = 1.96 \cdot 2.5738 \approx 5.0446\ 公斤/亩$$

$$75.6 - 5.0446 = 70.5554\ 公斤/亩$$

$$75.6 + 5.0446 = 80.6446\ 公斤/亩$$

该新型肥料对玉米产量增加量的总体均值的 95% 置信区间为(70.56 公斤/亩,80.64 公斤/亩)。